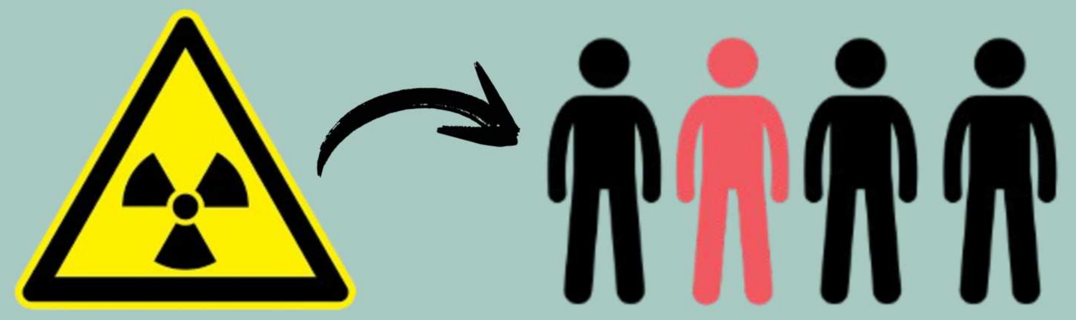
THE EFFECT OF MEDIORESINOL ON THE INDUCTION OF CELL DEATH OF BREAST CANCER CELLS THROUGH THE ACTIVATION OF THE P53 PROTEIN

CATEGORY: BIOMEDICAL & HEALTH SCIENCES

INTRODUCTION

- Treatment types
  - Not natural
  - Expensive
- In 2022 alone around 2.3 million women were diagnosed with breast cancer (World Health Organization: WHO & World Health Organization: WHO, 2024).



BACKGROUND

Medioresinol

Image taken by MuseuM Services Corporation, 2026

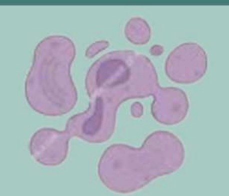
ROS + Oxidative Stress (HWANG ET AL., 2012)

Graphic made by researcher using Biorender, 2026

Oxidative Stress = p53 protein (LIU & XU, 2011)

Graphic made by researcher using Biorender, 2026

Cell Apoptosis



Graphic made by researcher using Biorender, 2026

HYPOTHESIS

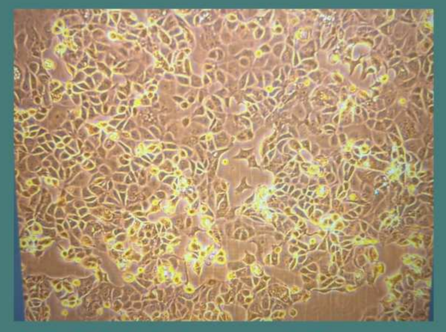
IF: MCF-7 (human breast cancer cells) are treated with medioresinol

THEN: There will be an increase in p53 protein expression.

JUSTIFICATION: Medioresinol has been shown to induce oxidative stress in cells which separately has been shown to increase p53 protein expression which has led to cell apoptosis.

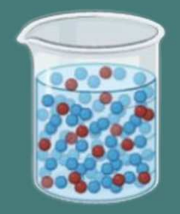
METHODOLOGY

CELL CULTURE + TREAT



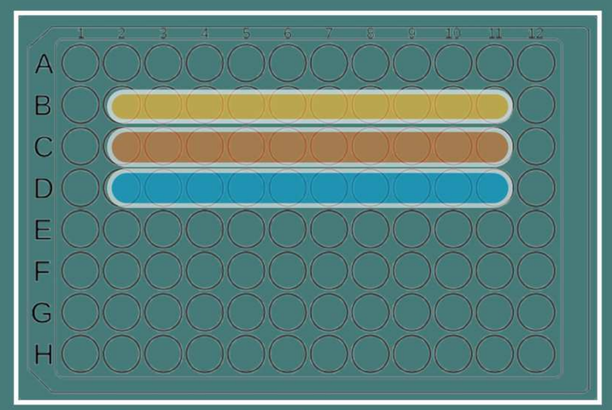
100X Magnification
90% Confluency

Images taken by researcher, 2026

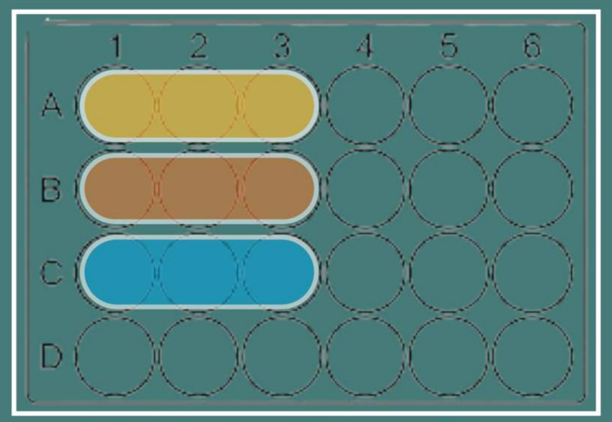


Medioresinol in DMSO

Graphic made by researcher using Biorender, 2026



96 well plate- Acid Phosphatase



24 well plate- Western Blot

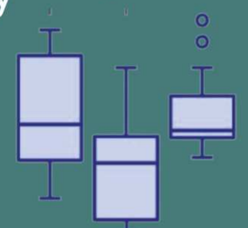
= 0μM- Negative Control
= 25μM-
= 50μM

Images made by researcher, 2026

IV: Medioresinol (25μM & 50μM)
DV: Upregulation of p53 protein

ACID PHOSPHATASE ASSAY

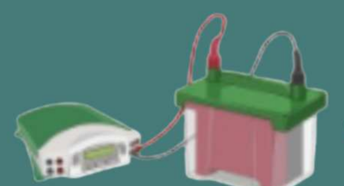
Test for cell viability



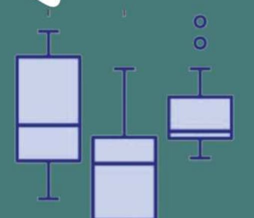
Kruskal-Wallis Test

Graphic made by researcher using Biorender, 2026

WESTERN BLOT



Test for p53 protein expression

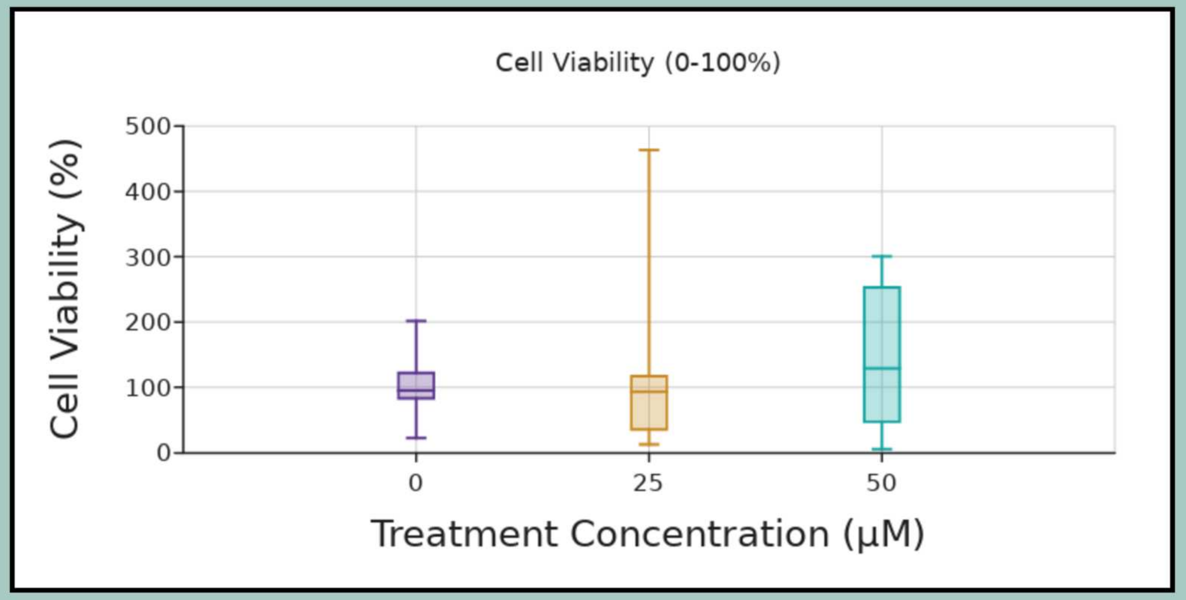


ANOVA Test

Graphic made by researcher using Biorender, 2026

RESULTS

Acid Phosphatase Assay:



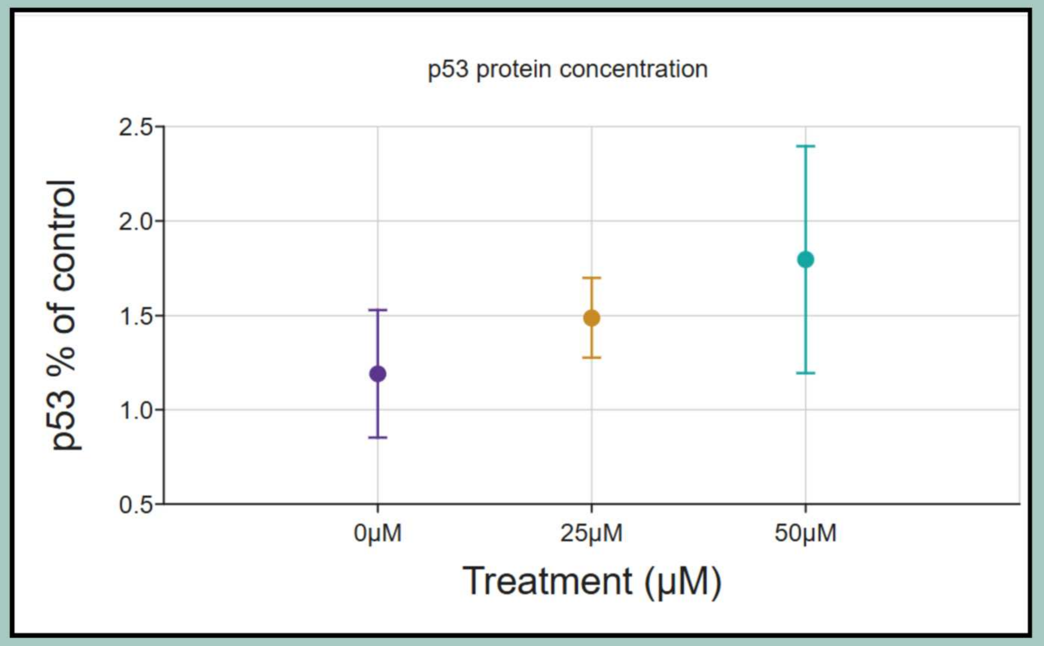
Data visualization created by researcher using DataClassroom, 2026

Kruskal Wallis Test:

Pairs with Significant Differences	Post Hoc P-Values	P-Value
0μM-25μM	1	0.08
0μM-50μM	0.23	
25μM- 50μM	0.12	

Table created by researcher using Google Sheets, 2026

Western Blot:



Data visualization created by researcher using DataClassroom, 2026

ANOVA Test:

Pairs with Significant Differences	Post Hoc P-Values	P-Value
0μM-25μM	0.65	0.23
0μM-50μM	0.21	
25μM- 50μM	0.58	

Table created by researcher using Google Sheets, 2026

DISCUSSION

Medioresinol did show an increase in p53 protein expression while an overall increase in cell viability was shown.

ACID PHOSPHATASE = CELL VIABILITY

- There was a slight decrease in cell viability when comparing the 25μM concentration to the control.
  - Conclusion: Medioresinol can decrease cell viability at a concentration near 25μM or lower.
- The overall p-value indicated that there was some confidence that the groups were different.

WESTERN BLOT = P53 PROTEIN

- The percent of p53 protein concentration increased as the medioresinol concentration increased.
  - Conclusion: Medioresinol can increase the p53 protein concentration in breast cancer cells.
- The overall p-value indicated that the was little confidence that the groups were different.

IMPLICATIONS:

- Medioresinol may be able to be used as a natural complementary treatment or potential treatment to kill breast cancer cells.
- Medioresinol can increase the p53 protein concentration in breast cancer cells.

FUTURE WORK:

- Test more specific dosages of medioresinol
- Test other lignans/flavinoids
- Measure downstream effectors of the p53 protein

REFERENCES

BioRender. (n.d.). *BioRender*. App.biorender.com. https://app.biorender.com/gallery

Hwang, J. H., Hwang, I., Liu, Q.-H., Woo, E.-R., & Lee, D. G. (2012a). (+)-Medioresinol leads to intracellular ROS accumulation and mitochondria-mediated apoptotic cell death in Candida albicans. *Biochimie*, *94*(8), 1784–1793. https://doi.org/10.1016/j.biochi.2012.04.010

Liu, D., & Xu, Y. (2011). p53, Oxidative Stress, and Aging. *Antioxidants & Redox Signaling*, *15*(6), 1669–1678. https://doi.org/10.1089/ars.2010.3644

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